Reading a genetic report
A published case: a VUS argued up to "likely pathogenic"

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Olivares-Huerta O, Castro-Coyotl DM, Crisanto-López IE, Cervantes-Larios J, Ochoa-Precoma R, Morales-López BF, Trejo-Toscano IA, Juárez-Melchor D. *Nabais Sá-de Vries Syndrome Type 1 in a Mexican Girl: A Case Report.* Cureus. 2026;18(4):e107064. doi:10.7759/cureus.107064. Medical Genetics, IMSS General Hospital Zone 20, Puebla, Mexico, and partner institutions. cureus.com/articles/478396

The patient

- Girl, now 5 years old; second twin of a dizygotic pregnancy conceived by assisted reproduction
- Respiratory distress in the first days; echocardiography: patent ductus arteriosus, patent foramen ovale, pulmonary hypertension
- Distinct facial features, hemangioma, congenital melanocytic nevus, inguinal hernia
- Microcephaly, short stature, low weight; severe developmental and language delay; hearing normal

Work-up

- Genetics review from 1 week of age (congenital heart disease)
- Karyotype: 46,XX (normal)
- Whole exome sequencing for the cardiac and neurological findings
- Result: heterozygous missense variant **SPOP c.351G>T, p.(Met117Ile)** (NM_001007228.2), in the MATH domain, reported as a **variant of uncertain significance**

How the authors re-assessed the variant (ACMG/AMP criteria)

Code	Strength	Evidence in this case
PM2	Moderate	Absent / extremely rare in the population database (gnomAD)
PM1	Moderate	Located in a critical functional domain of SPOP (MATH domain)
PM6	Moderate	De novo *assumed* from the pedigree, **not confirmed** by parental testing
PP3	Supporting	Computational (in silico) tools predict a damaging effect

On this basis the authors advocate reclassifying the variant as **likely pathogenic** and report the diagnosis as Nabais Sá-de Vries syndrome type 1. They note that no functional assay exists for this variant and that **no segregation analysis** was done.

Talking points for the doctor visit

- A VUS is not a diagnosis: do not base treatment or relatives' predictive testing on it.
- The phenotype, the gene's known disease and the inheritance pattern must fit together.
- Testing both parents is often the simplest way to resolve a VUS: a **confirmed** de novo variant counts as strong evidence (PS2) instead of moderate (PM6).
- Classifications change; ask the laboratory for re-analysis or re-classification over time.

For healthcare professionals' general awareness. Not a substitute for clinical judgement; facts and guidelines may change. Case details are as published by the original authors; the classification shown is the authors' assessment. Consult the full article.

EPISODE 2 · QUICK GUIDE

How to read a genetic report

Report item	What to check
Indication & test	Was the right test done for the clinical question? Were full clinical details sent? Proband only or trio?
Variant (HGVS)	c. = coding DNA change (e.g. c.351G>T: G replaced by T at position 351); p. = protein change (e.g. p.Met117Ile: methionine to isoleucine at amino acid 117).
Zygosity	Heterozygous (one copy), homozygous (both copies), hemizygous (X-linked in males). In a recessive disease, one heterozygous variant usually means a carrier.
Classification	Pathogenic · Likely pathogenic (>90% certainty) · VUS · Likely benign · Benign (ACMG/AMP 2015).
Phenotype fit	Does the gene's known disease match the patient? A pathogenic variant that does not fit may not explain the current illness.
Secondary findings	Actionable variants unrelated to the indication (ACMG SF list, 80+ genes; v3.3, 2025). Reported only as agreed at consent.
Limitations	A negative exome does not exclude a genetic disease (repeat expansions, many CNVs, mitochondrial DNA may be missed). Re-analysis after 1–2 years can help.

Can a VUS change?

In about 1.45 million people tested for hereditary cancer (2006–2016), 24.9% of reported VUS were reclassified: 91.2% to benign/likely benign and 8.7% to pathogenic/likely pathogenic. Median time about 1.2 years (down) and 1.9 years (up).

VUS: do and don't

Don't: treat, or test relatives predictively, on a VUS.
Do: correlate with features, test parents/relatives if advised, ask for re-analysis, and offer genetic counselling.

References

- Olivares-Huerta O et al. Cureus 2026;18(4):e107064. cureus.com/articles/478396 · PubMed 42147571
- Richards S et al. Standards and guidelines for the interpretation of sequence variants (ACMG/AMP). Genet Med 2015;17(5):405–424.
- Mersch J, Ross TS et al. Prevalence of variant reclassification following hereditary cancer genetic testing. JAMA 2018; summary: Healio, Dec 2018
- ACMG SF v3.3 secondary findings gene list (2025). acmg.net press release

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